

AI Assisted Coding

Assignment 4.5

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Task-01:

Prompt:

Write a Python program that defines a function to determine whether a given year is a leap year. The function should take the year as user input and display whether it is a leap year or not.

Code:

```
# Define a function that determines whether a given year is a leap year.
def is_leap_year(year):
    # Apply the standard leap year conditions:
    # 1) Year must be divisible by 4
    # 2) If divisible by 100, it must also be divisible by 400
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        return True
    # Otherwise, return False
    return False

# Ask the user to enter a year
year = int(input("Enter a year: "))

# Call the function and display the result clearly
if is_leap_year(year):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")

# Display a formatted comparison table for explanation purposes
print("\n" + "=" * 60)
print("COMPARISON TABLE")
print("=" * 60)

print("""
| Readability | Verbose | Concise, Pythonic |
|-----|-----|-----|
| Large Inputs | Slow, inefficient | Fast, optimized |
| Educational | Great for learning | Practical, modern |
| When to Use | Learning/Teaching | Production code |
| | Understanding basics | Real-world apps |
| | | Performance-critical |
""")
```

Output:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter a year: 2005
2005 is not a leap year.
```

COMPARISON TABLE

Readability	Verbose	Concise, Pythonic
Large Inputs	Slow, inefficient	Fast, optimized
Educational	Great for learning	Practical, modern
When to Use	Learning/Teaching	Production code
	Understanding basics	Real-world apps
		Performance-critical

Task02:

Prompt:

write a python code for converting centimetres to inches by using mathematical formula. Example when user gave input 10cm and output should be 3.94 inches

CODE:

```
# Create a function that converts centimetres to inches using the formula:
# inches = centimetres / 2.54
def cm_to_inches(cm):
    inches = cm / 2.54
    return inches

# Ask the user to enter the length in centimetres
cm_value = float(input("Enter length in centimetres: "))

# Call the conversion function and store the result
inches_value = cm_to_inches(cm_value)

# Display the result rounded to two decimal places
print(f"{cm_value} cm is equal to {inches_value:.2f} inches.")
```

Output:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter length in centimetres: 10
10.0 cm is equal to 3.94 inches.
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter length in centimetres: 13
13.0 cm is equal to 5.12 inches.
```

Task_03

Prompt: Write a Python program that defines a function to format a person's name.

The function should take first name and last name as input from the user and return the formatted result in the form: "Last Name, First Name".

Test the function with examples like:

First name: Rahul, Last name: Verma

First name: Priya, Last name: Nair

CODE:

```
# Define a function that formats the name in "Last Name, First Name" format.
# Ensure both names are properly capitalized and stripped of extra spaces.
def format_name(first_name, last_name):
    first_name = first_name.strip().capitalize()
    last_name = last_name.strip().capitalize()
    return f"{last_name}, {first_name}"

# Take user input for first and last name.
first_name = input("Enter first name: ")
last_name = input("Enter last name: ")

# Call the function and display the formatted result.
formatted_name = format_name(first_name, last_name)
print(f"Formatted Name: {formatted_name}")
```

OUTPUT:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter first name: Nihaal
Enter last name: Dasavaram
Formatted Name: Dasavaram, Nihaal
```

Task_04

Prompt: # Write a Python function that accepts a string from the user and calculates how many vowels are present in it.

The function should count both uppercase and lowercase vowels.

Code:

```
# Create a function that counts vowels in a given string.
# Ensure both uppercase and lowercase vowels are considered.
def count_vowels(input_string):

    # Define all vowels explicitly.
    vowels = "aeiouAEIOU"

    # Use a generator expression to efficiently count matching characters.
    count = sum(1 for char in input_string if char in vowels)

    # Return the total count.
    return count

# Prompt the user to enter a string.
user_string = input("Enter a string: ")
# Call the function and print the result clearly.
vowel_count = count_vowels(user_string)
print(f"Number of vowels in the string: {vowel_count}")
```

OUTPUT:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter a string: levi
Number of vowels in the string: 2
```

Few Examples:

Example 1:

Input: "ackerman" Output: 3

Example 2:

Input: "mikasa" Output: 3

Now write a Python function that takes a string as input and returns the number of vowels in it. Count both uppercase and lowercase vowels.

Output:

```
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter a string: ackerman
Number of vowels in the string: 3
PS C:\Users\nihaa> & C:/Users/nihaa/AppData/Local/Programs/Python/Python313/python.exe
Enter a string: mikasa
Number of vowels in the string: 3
PS C:\Users\nihaa> |
```

Task_05:

'''Write a Python function that opens a text file (.txt), reads its contents, counts how many lines are present, and returns the total number of lines. Test the function using different sample text files.'''

CODE:

```
# Define a function that takes a file name as input.
# The function should open the file, read all lines,
# count them, and return the total number of lines.
def count_lines(file_name):

    # Open the file in read mode.
    with open(file_name, "r") as file:

        # Read all lines from the file.
        lines = file.readlines()

        # Return the total number of lines.
        return len(lines)

# Test the function with sample files.
file1 = "sample1.txt"
file2 = "sample2.txt"

print(f"Number of lines in {file1} = {count_lines(file1)}")
print(f"Number of lines in {file2} = {count_lines(file2)}")
|
```